EPS 112—Caldor Fire Field Trip Report
CORRIGENDUM

Note 1: In the erosion pin tables, pin heights are in mm, not cm.

Note 2: In sections 1 and 2, don't over-interpret signals for each individual pins. Rather, focus on temporal trends for each site as a whole, and any overall spatial trends within each site.

3. Sediment Transport on Hillslopes Across Timescales

3.1. How do site-specific long-term erosion rates (provided in 2.2) compare with short-term rates of erosion/accumulation over the past 5 years (calculated in 1.3)? Why?

3.2. Using the long-term erosion rate (E) for each site, predict the <u>steady-state</u> (average) soil depth at Sly Park (P_0 = 66 m/Myr, h* = 0.5 m) and Capp's Crossing (P_0 = 44 m/Myr, h* = 0.5 m). How do predictions compare with field observations? Hint: at steady state, you can assume that the soil-production rate, P = E.

3.3. Estimate the soil residence timescales at both sites. Hint: The residence timescale is the time required to erode one full soil thickness.

3.4. The andesite_ridge_soil_thickness_curvature attribute table provides your field measurements of soil thickness. Use them to calculate the residence timescale of soil along those slopes. Do these make sense, and why?

3.5. For the andesite ridges, make a plot of soil thickness vs. elevation. Does it make sense, and why? Calculate the soil production rate, P, in m/Myr using P_0 = 36 m/Myr and h* = 0.5 m. Generate a plot of P = f(soil thickness), with the y-axis in log scale. How do these compare with soil production rates measured by Heimsath (1997) at Tennessee Valley (see plots in lecture slides)?